

Course Code	Name of the Course				
216U03L504	Operating Systems Laboratory				
Teaching Scheme	TH	P	TUT	Total	
	--	02		02	
Credits Assigned	--	01		01	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	50	--	--	--	50

Course pre-requisites: Nil

Course Objectives: This course gives general description of the basic concepts of operating systems, functionalities and objectives, structure and components. It helps in understanding the services provided by and the design of an operating system. It helps to understand what a process is and how processes are synchronized and scheduled. The course covers the structure and organization of the file system, memory management, I/O management.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO 1.** Understand the role of an operating system, its function and issues.
- CO 2.** Compare different algorithms used for management and scheduling of processes.
- CO 3:** Understand different approaches to memory management.
- CO 4:** Understand the structure and organization of the file system.
- CO 5:** Explore I/O systems, Storage structure and Unix system.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Fundamental of Operating System(OS)			
	1.1	Definition, objectives, functions, services, types, and different views of OS.	04	CO1
	1.2	Operating System as a resource manager, system calls, and types of shells. Monolithic systems, layered systems, client server model, monolithic kernel and microkernel.		
		#Self-Study: Comparison of different types of operating systems in terms of OS structures.		
2	Process Management			
	2.1	Different states of a Process, Process State transition diagram. Thread: Definition, Various states, Types of threads, Concept of multithreads. Inter-process Communication: Concurrent processes, Critical Section, Race Conditions, Mutual Exclusion, Hardware Solution, Semaphores, Peterson's Solution, The Producer / Consumer Problem.	10	CO2
	2.2	Deadlocks: Definition, Necessary and sufficient conditions for Deadlock, Deadlock Prevention, and Deadlock Avoidance: Banker's algorithm, Deadlock detection and Recovery.		
	2.3	Types of Schedulers, Scheduling criteria: CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time. Scheduling algorithms: Pre-emptive and non-pre-emptive, FCFS, SJF, Round-robin.		
		#Self-Study: Implementation of different scheduling algorithms using Shell script programs. Comparing types of process scheduling algorithms in terms of minimum Turn-around time, Waiting time and response time. Implement The Producer / Consumer Problem, Banker's algorithm in Linux/Unix shell scripts.		
3	Memory Management			
	3.1	Basic concept, Logical and Physical address maps, Memory allocation strategies fragmentation and Compaction.	06	CO3
	3.2	Basics of Virtual Memory, Page allocation, Partitioning, Paging, Page fault, Segmentation, Demand paging, Page Replacement algorithms: Optimal, First in First Out (FIFO) and Least Recently used (LRU).		

4	File Management		06	CO4
	4.1	Concept of File, Access methods, File types, File operation, Directory structure, File System structure.		
	4.2	File allocation methods (contiguous, linked, indexed), Free-space management (bit vector, linked list, grouping), directory implementation. #Self-Study: Implementation of File allocation methods using Linux/Unix shell scripts.		
5	Storage and I/O management		04	CO5
	5.1	I/O Hardware: I/O devices, Device controllers, Direct Memory Access, Principles of I/O. Disk Management: Disk structure, Disk scheduling - FCFS, SSTF, SCAN, C-SCAN.		
	5.2	#Self-Study: Implementation of File allocation methods using Linux/Unix shell scripts.		
		Total	30	

- LAB CA includes journal / written record of experiments based on above contents.
- LAB CA also includes evaluations covering entire syllabus
- LAB CA schedule, evaluation scheme and detailed rubrics will be declared at the starting of the semester by the concerned faculty in-charge.

Reference Books*

Sr. No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Silberschatz A., Galvin P., and Gagne G	<i>Operating Systems Concepts</i>	Wiley Eastern Limited, India	9 th Edition, 2018
2	William Stallings,	<i>Operating System- Internal & Design Principles</i>	Pearson, London, England	7 th Edition, 2013
3	Maurice J. Bach	<i>The Design of Unix Operating System</i>	Prentice Hall, United States	1 st Edition 2015
4	Andrew S. Tanenbaum. Herbert Bos	<i>Modern Operating Systems</i>	PHI Learning Pvt. Ltd, India	4 th Edition 2016